

Baseline Verification — Default vs. Joint

4DVarNet-FM: DA Baselines – Default vs. Joint Parameter Estimation

Common parameters:

T_max = 3.0s dt = 0.01 Steps = 300
obs_interval = 20 (15 obs / window, includes step 0) R_var = 0.5 B_var = 2.0
DA window steps (DWS) = 50

Method parameters (shared by Default and Joint variants):

Weak-4DVar opt_steps=150, lr=0.02, B_var=2.0, Q_var=0.05
Strong-4DVar max_iter=40, lr=0.1, B_var=2.0
EnKF N_ensemble=30, inflation=2.0
ETKF N_ensemble=30, inflation=2.0 (deterministic transform)

Default vs. Joint:

Default : solves with fixed, nominal sigma/rho/beta/c1.
Joint : additionally estimates sigma/rho/beta/c1 online (log-space,
Gaussian prior P_var=1.0 around the DA-side nominal values,
re-estimated every DA window and carried forward).

Case studies:

Parameter	S0	S1
Truth coupling exponent	1.6	1.6
DA coupling exponent	1.6 (perfect model)	1.0 (mismatch)
Param bias	0.0	0.15
Forcing-state bias	0.0	0.1

Test data:

200 test windows per case, RandomParamLorenz63Dataset (per-window random
sigma/rho/beta +/-20%), make_s0_s1_trainval data setup.

Source: experiments/baselines_joint_dws50_s0s1_inf2.0_etkf_inf2.0.json

Total run time: 470s

Baseline Verification — Metrics (Default vs. Joint)

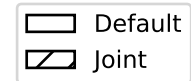
Summary of Results — Mean State RMSE ± std, Default vs. Joint

Method	S0 Default	S0 Joint	S0 dJ%	S1 Default	S1 Joint	S1 dJ%	Deg (Def)	Deg (Joint)
Weak-4DVar	0.6386±0.4999	0.9767±0.4982	+52.9%	1.6341±0.7591	0.9586±0.3864	-41.3%	2.56x	0.98x
Strong-4DVar	0.7228±0.9094	0.9973±0.7517	+38.0%	2.1249±1.1546	0.9937±0.7835	-53.2%	2.94x	1.00x
EnKF	0.7792±0.1724	1.3447±0.4239	+72.6%	2.2765±0.4201	1.4628±0.4075	-35.7%	2.92x	1.09x
ETKF	0.7719±0.1708	0.9720±0.5108	+25.9%	2.2842±0.5087	1.4971±0.6793	-34.5%	2.96x	1.54x

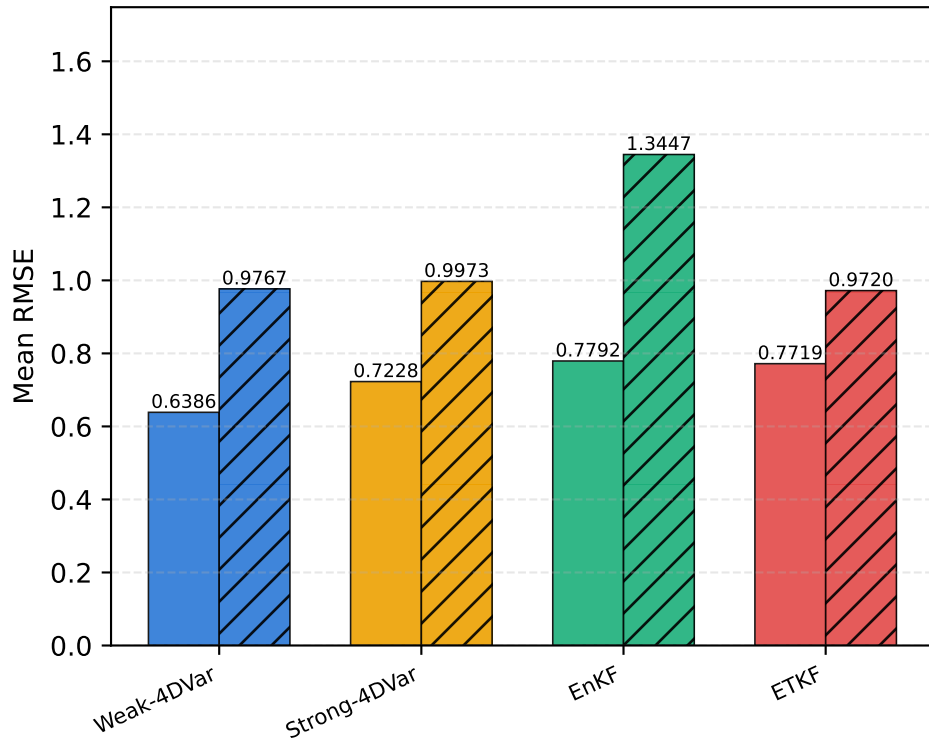
dJ% = (Joint mean - Default mean) / Default mean. Positive = joint estimation costs state accuracy.
Deg = S1 mean / S0 mean (degradation under model mismatch). Lower = more robust.

Best Default on S0: Weak-4DVar (0.6386)
Best Joint on S0: ETKF (0.9720)
Best Default on S1: Weak-4DVar (1.6341)
Best Joint on S1: Weak-4DVar (0.9586)

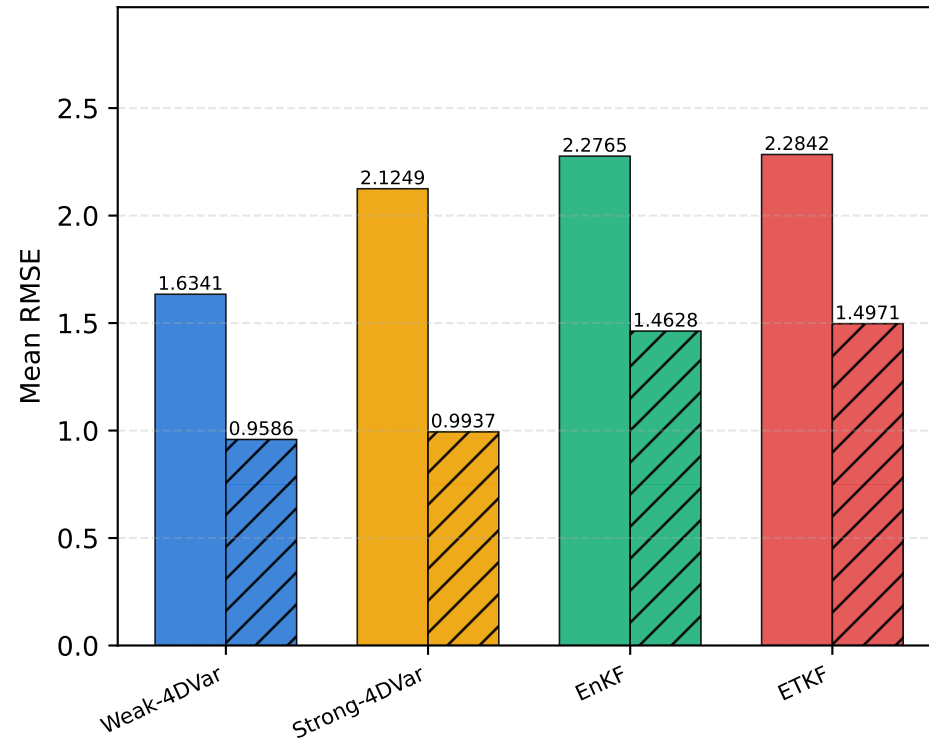
Baseline Comparison — Default vs. Joint



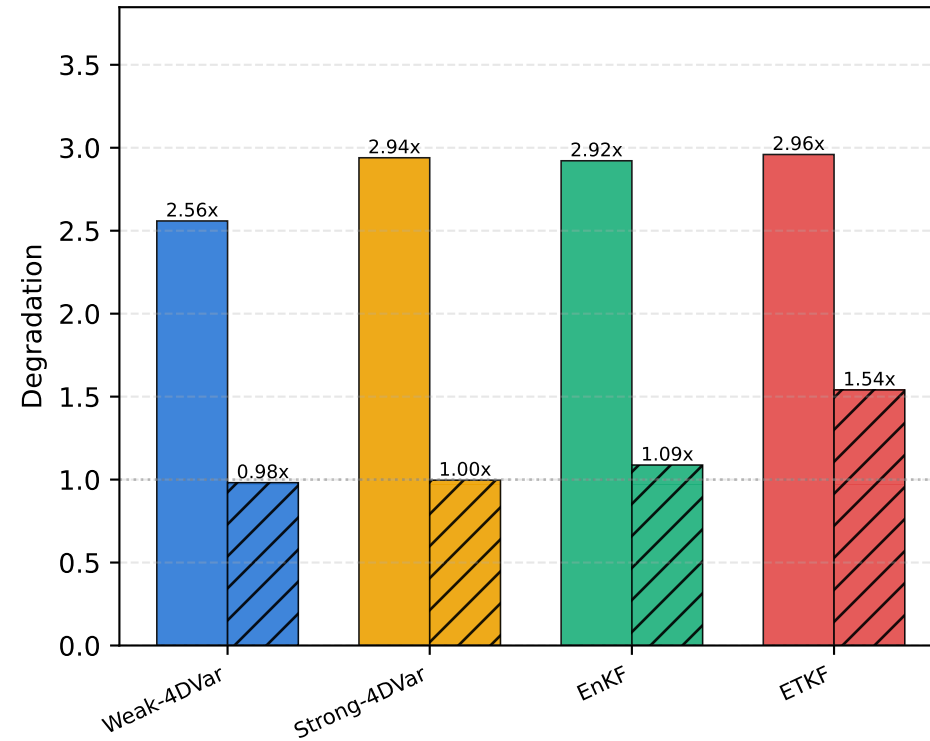
S0 Mean RMSE



S1 Mean RMSE



Robustness (S1/S0)



Per-Component RMSE Breakdown

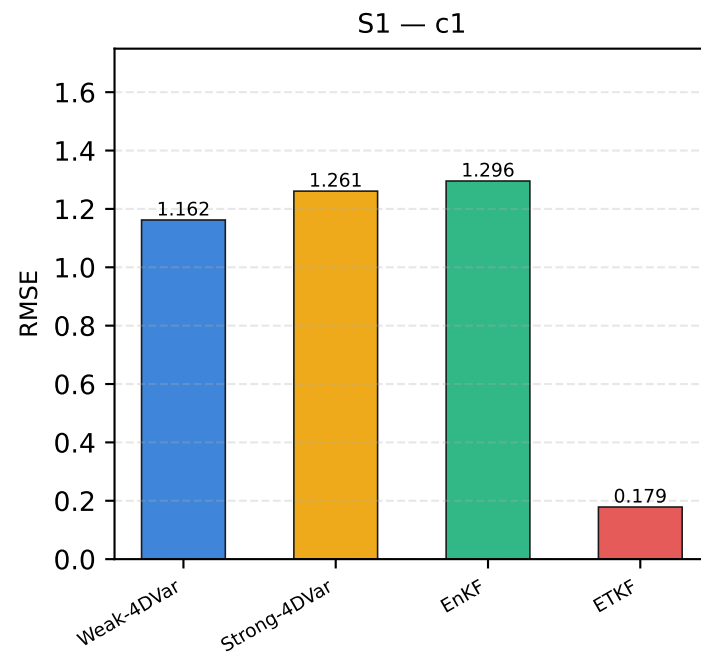
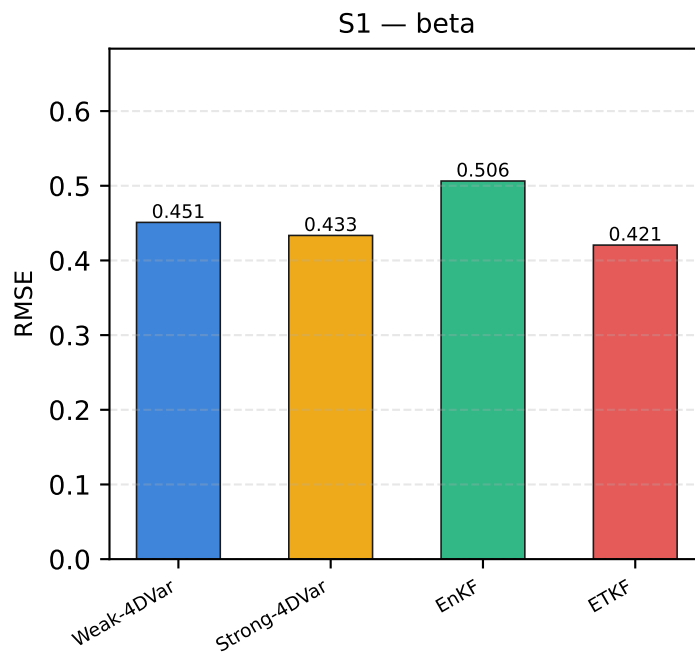
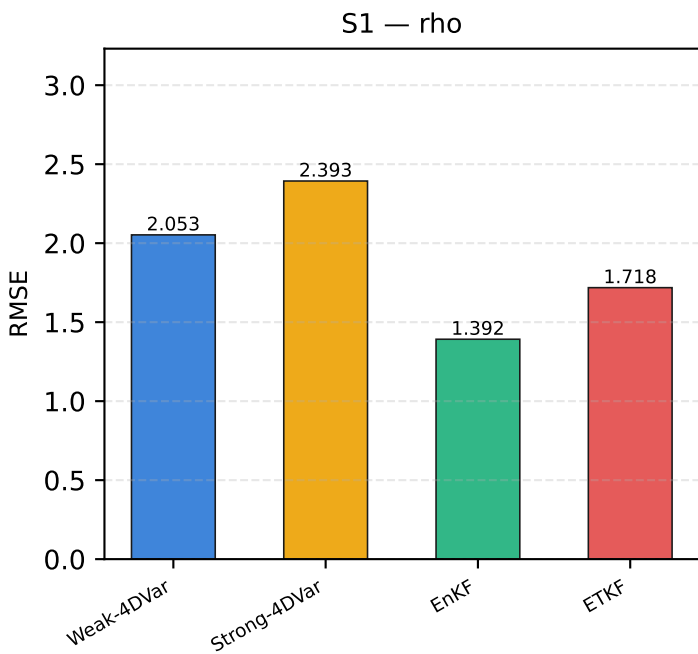
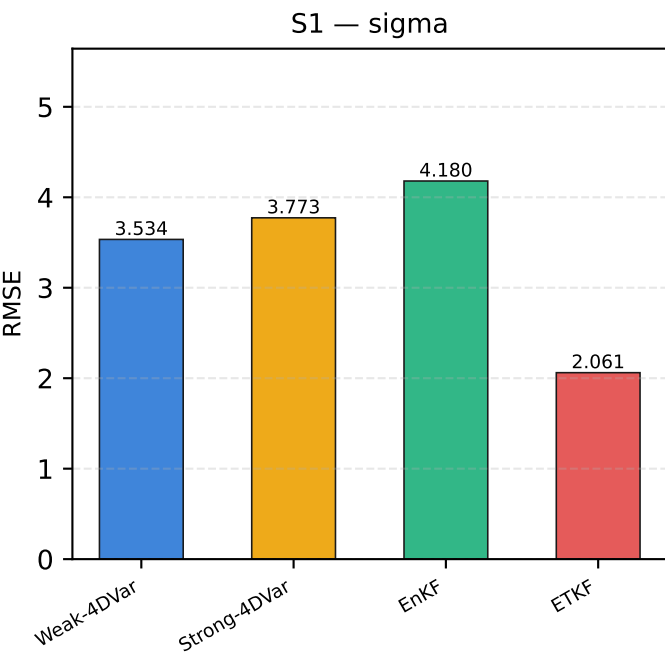
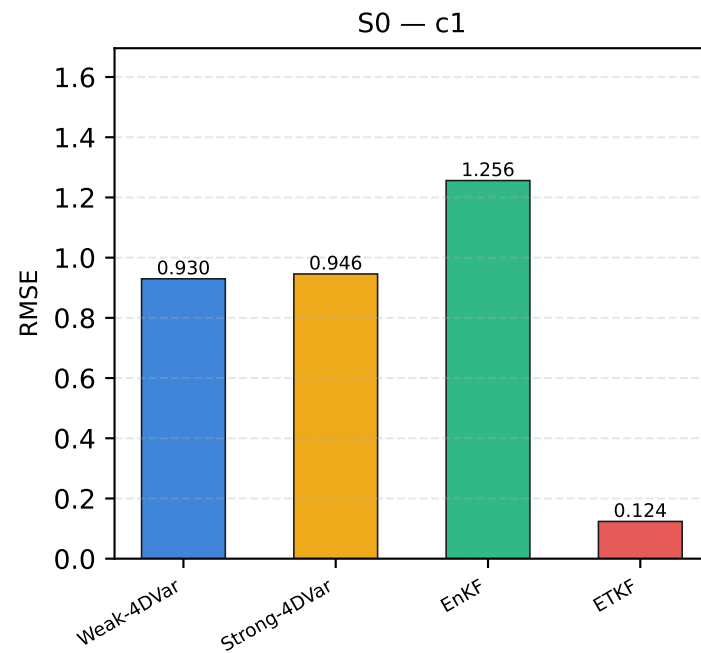
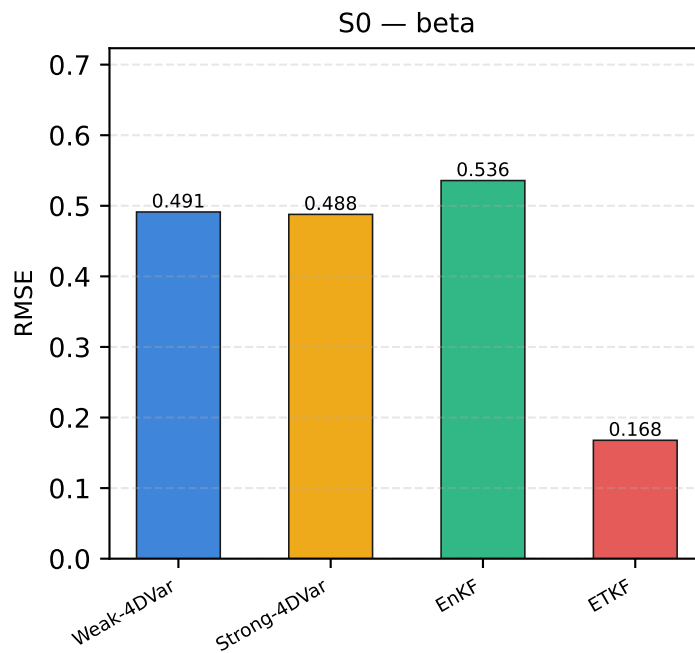
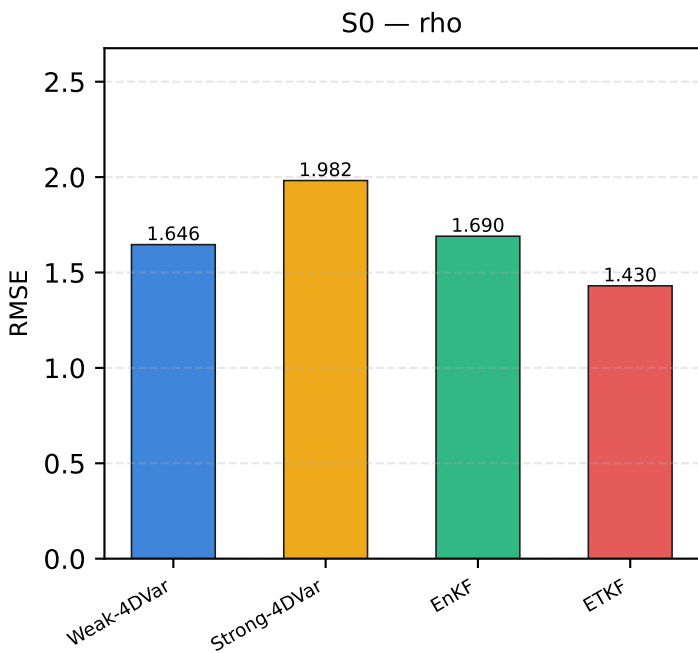
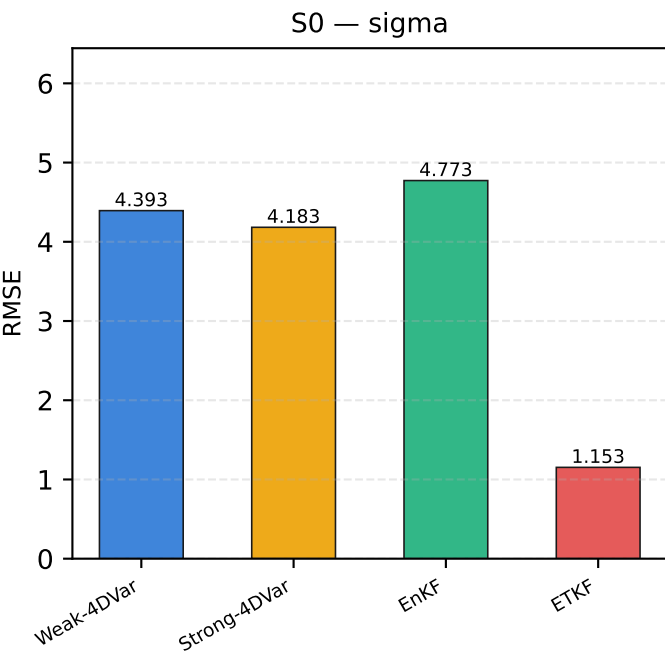
Per-Component RMSE ± std – Default vs. Joint

----- S0 -----					
Method	Variant	X	Y	Z	Mean

Weak-4DVar	Default	0.4495±0.4259	0.6579±0.4482	0.8085±0.6061	0.6386±0.4999
Weak-4DVar	Joint	0.7403±0.3651	0.9858±0.5363	1.2039±0.5690	0.9767±0.4982
Strong-4DVar	Default	0.5108±0.7081	0.7208±0.8750	0.9368±1.1019	0.7228±0.9094
Strong-4DVar	Joint	0.8091±0.7299	0.9978±0.8325	1.1850±0.6853	0.9973±0.7517
EnKF	Default	0.5275±0.1148	0.8572±0.2121	0.9529±0.1760	0.7792±0.1724
EnKF	Joint	1.1713±0.4016	1.3490±0.4419	1.5140±0.4272	1.3447±0.4239
ETKF	Default	0.5209±0.1128	0.8473±0.2029	0.9473±0.1833	0.7719±0.1708
ETKF	Joint	0.6422±0.3223	1.0246±0.5846	1.2491±0.5806	0.9720±0.5108
----- S1 -----					
Method	Variant	X	Y	Z	Mean

Weak-4DVar	Default	0.8283±0.7344	1.3046±0.9115	2.7695±0.5990	1.6341±0.7591
Weak-4DVar	Joint	0.7395±0.2580	0.9767±0.4437	1.1595±0.4296	0.9586±0.3864
Strong-4DVar	Default	1.1144±1.1534	1.5827±1.3919	3.6776±0.8554	2.1249±1.1546
Strong-4DVar	Joint	0.8017±0.6890	0.9960±0.8149	1.1835±0.8382	0.9937±0.7835
EnKF	Default	1.1233±0.2798	1.9655±0.3749	3.7406±0.5572	2.2765±0.4201
EnKF	Joint	1.2321±0.3950	1.4575±0.4299	1.6987±0.3967	1.4628±0.4075
ETKF	Default	1.1251±0.3588	2.0235±0.5508	3.7040±0.5866	2.2842±0.5087
ETKF	Joint	1.0540±0.4994	1.5108±0.6954	1.9267±0.8070	1.4971±0.6793

Parameter RMSE Breakdown — Joint Variants (S0 vs. S1)



Parameter RMSE Table — Joint Variants

Parameter RMSE $\pm$ std — Joint Variants Only

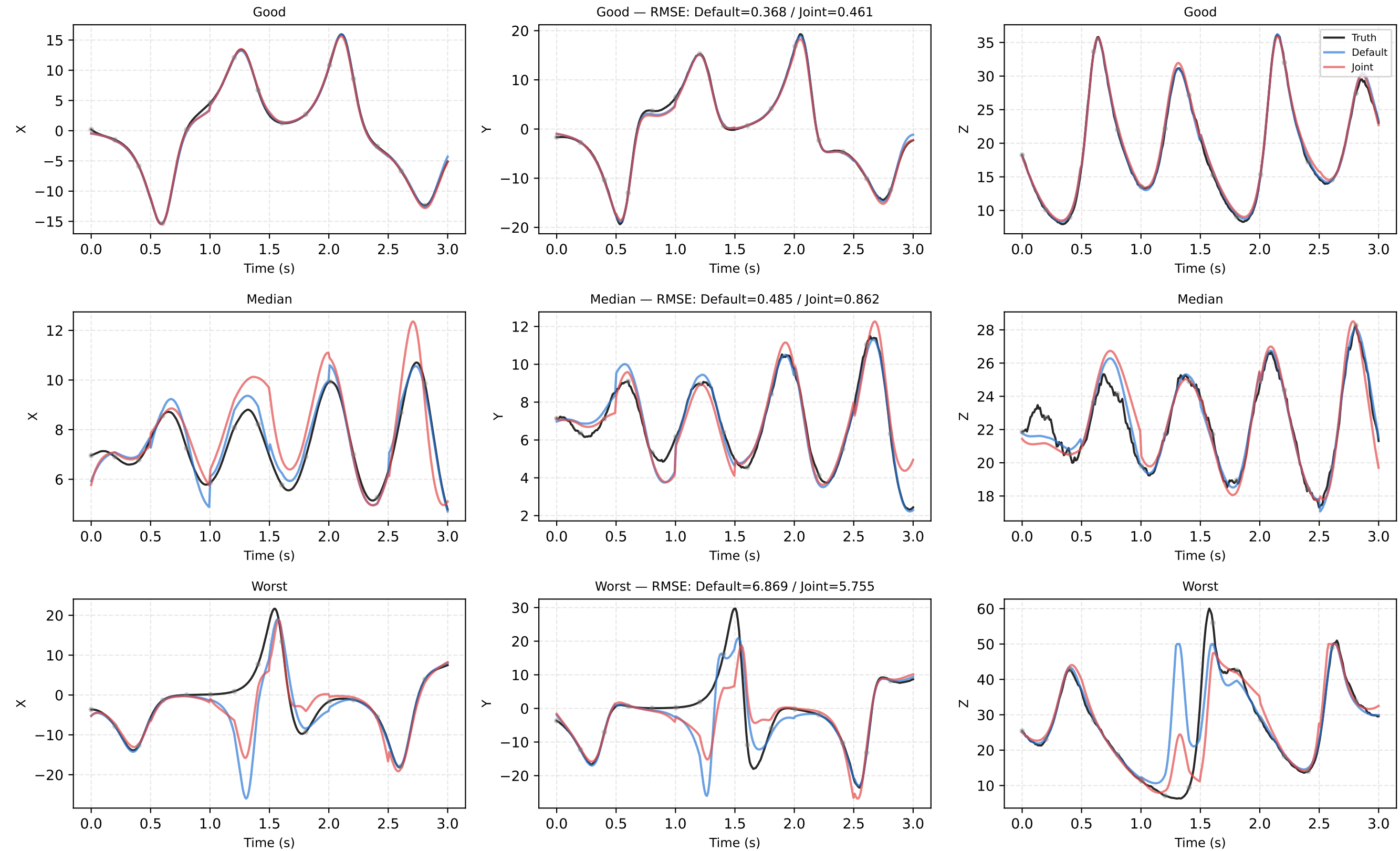
--- S0 ---

Method	sigma	rho	beta	c1
Weak-4DVar	4.3932 $\pm$ 3.5084	1.6461 $\pm$ 1.2557	0.4912 $\pm$ 0.3557	0.9297 $\pm$ 0.7640
Strong-4DVar	4.1830 $\pm$ 3.2831	1.9819 $\pm$ 1.6022	0.4878 $\pm$ 0.3399	0.9458 $\pm$ 0.7712
EnKF	4.7729 $\pm$ 2.4826	1.6902 $\pm$ 1.2540	0.5358 $\pm$ 0.3447	1.2561 $\pm$ 0.3410
ETKF	1.1528 $\pm$ 0.9811	1.4302 $\pm$ 1.1920	0.1678 $\pm$ 0.1087	0.1237 $\pm$ 0.1185

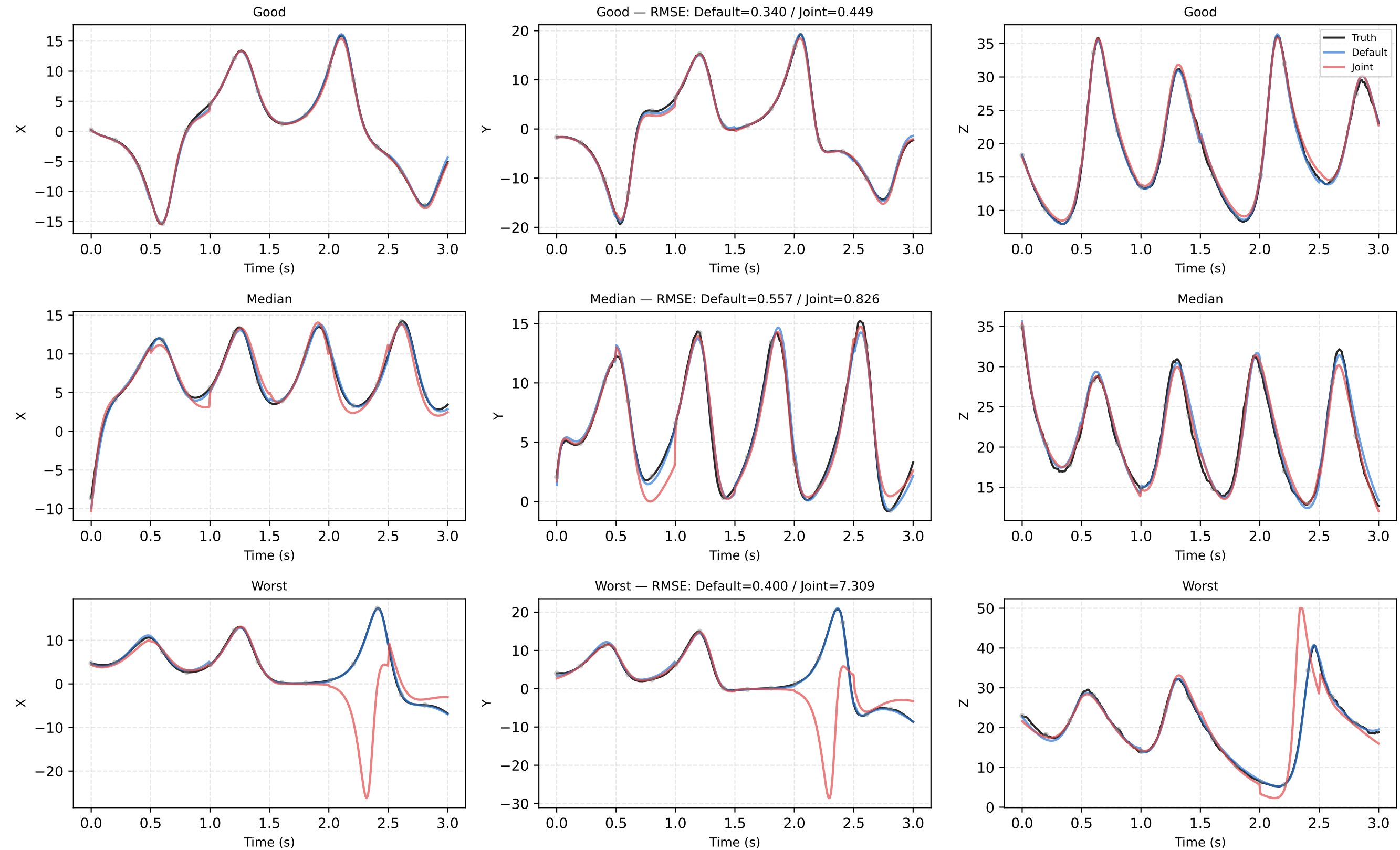
--- S1 ---

Method	sigma	rho	beta	c1
Weak-4DVar	3.5342 $\pm$ 2.6452	2.0525 $\pm$ 1.6773	0.4510 $\pm$ 0.2971	1.1621 $\pm$ 0.9210
Strong-4DVar	3.7730 $\pm$ 2.7660	2.3935 $\pm$ 2.0549	0.4335 $\pm$ 0.2829	1.2610 $\pm$ 0.9992
EnKF	4.1798 $\pm$ 2.2296	1.3919 $\pm$ 0.9875	0.5064 $\pm$ 0.3513	1.2956 $\pm$ 0.3542
ETKF	2.0612 $\pm$ 1.1192	1.7184 $\pm$ 1.2371	0.4206 $\pm$ 0.3046	0.1786 $\pm$ 0.0844

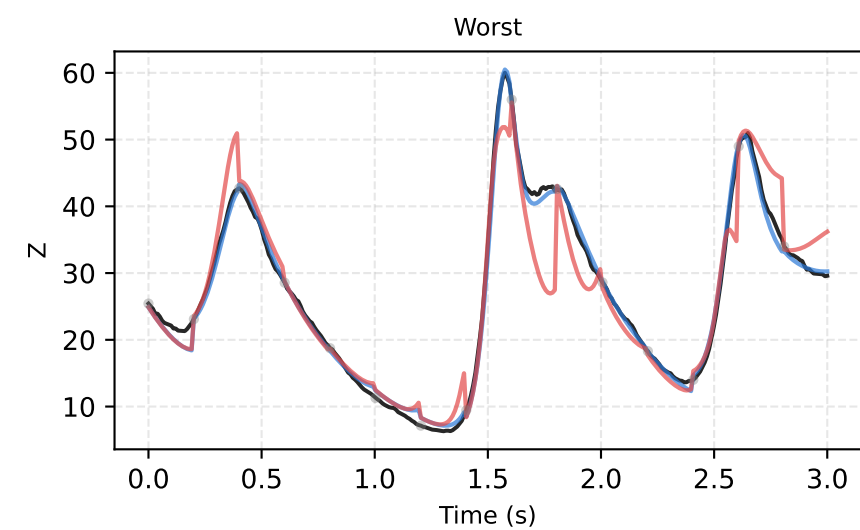
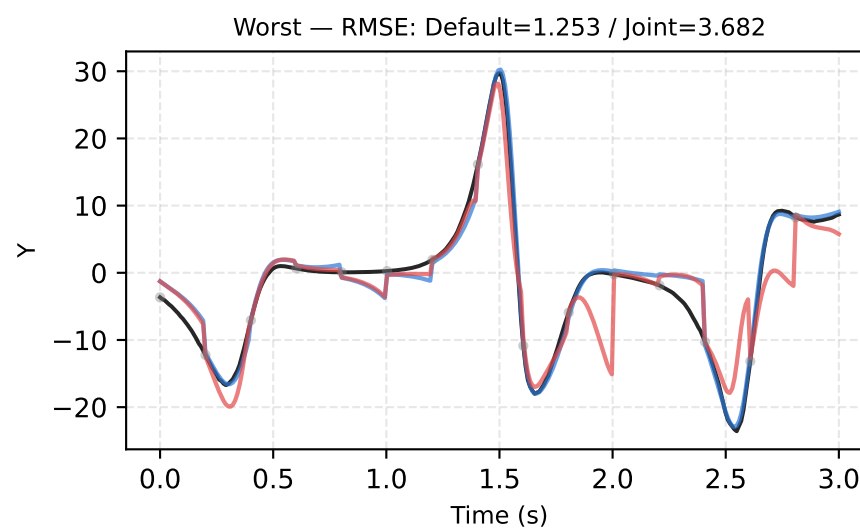
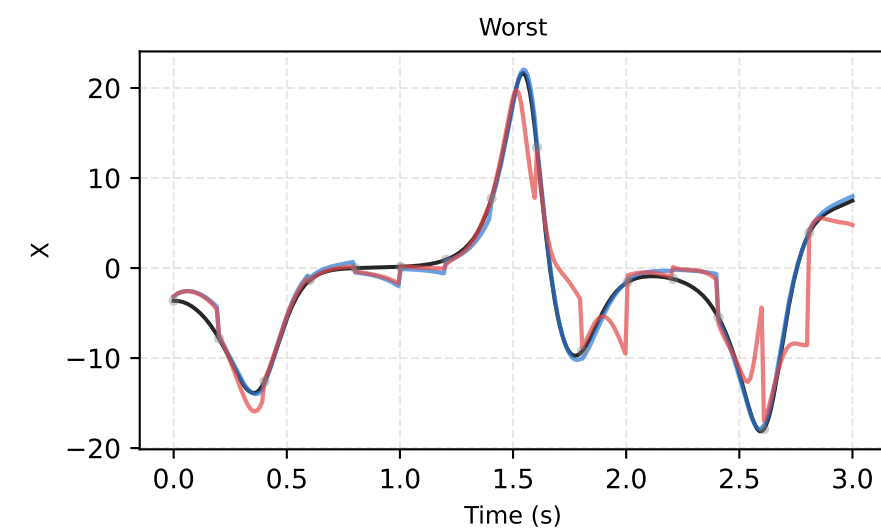
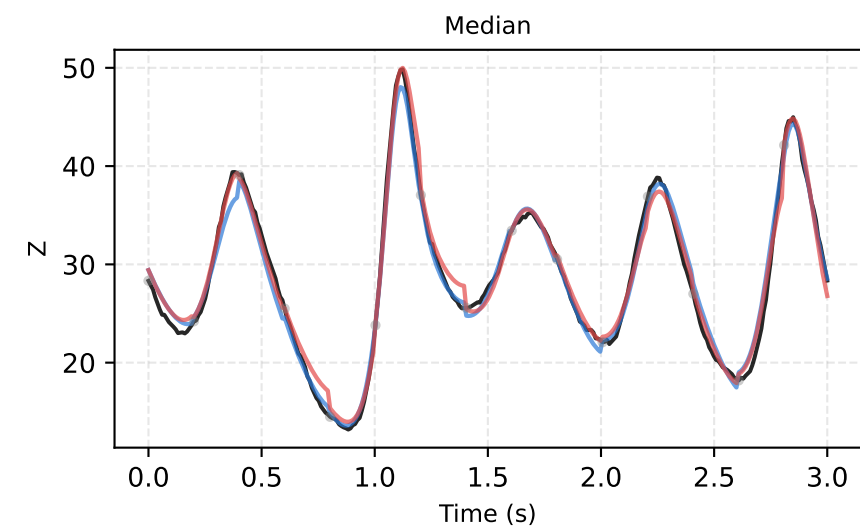
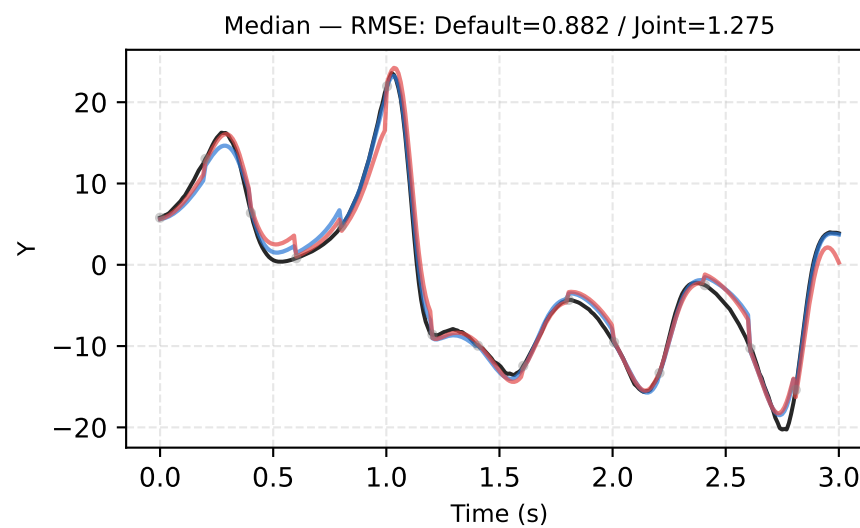
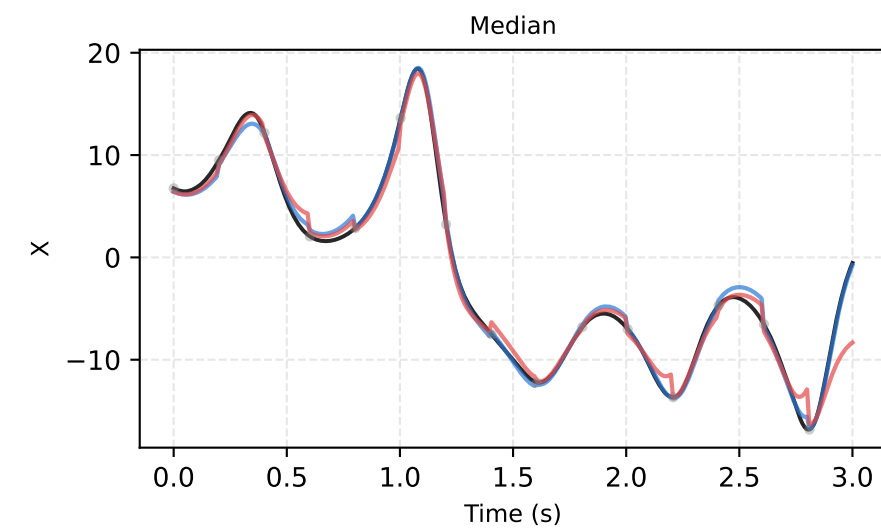
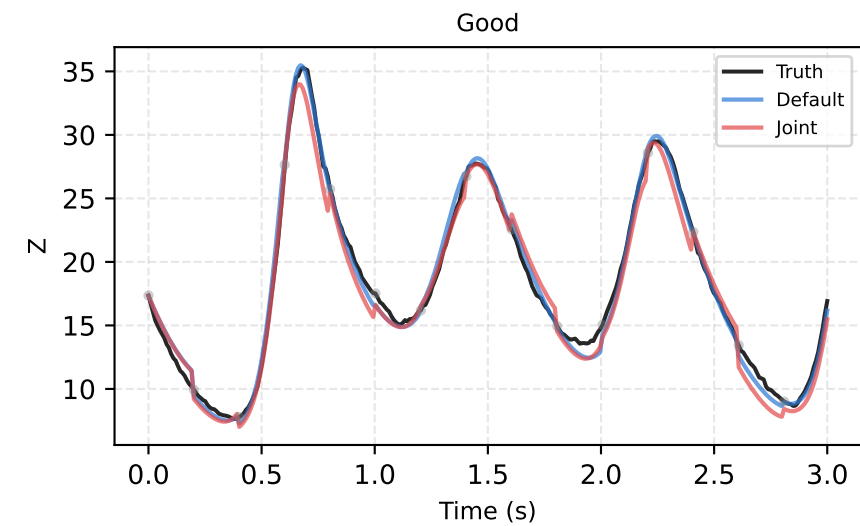
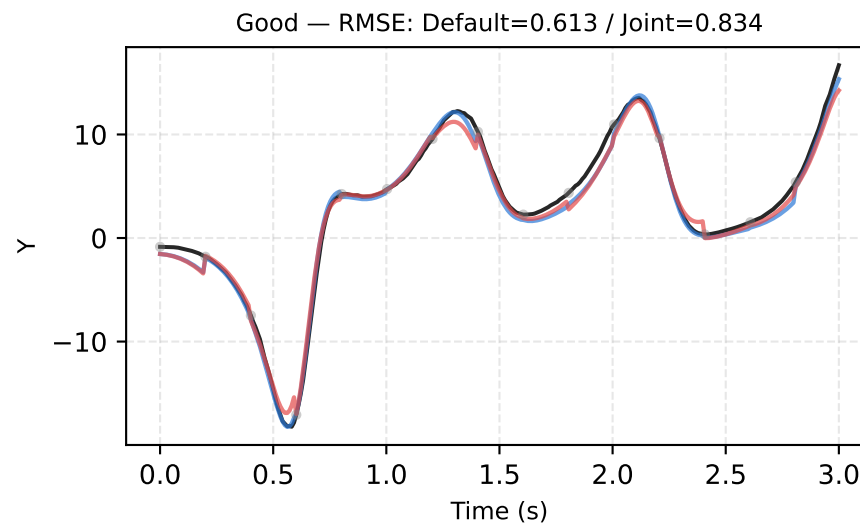
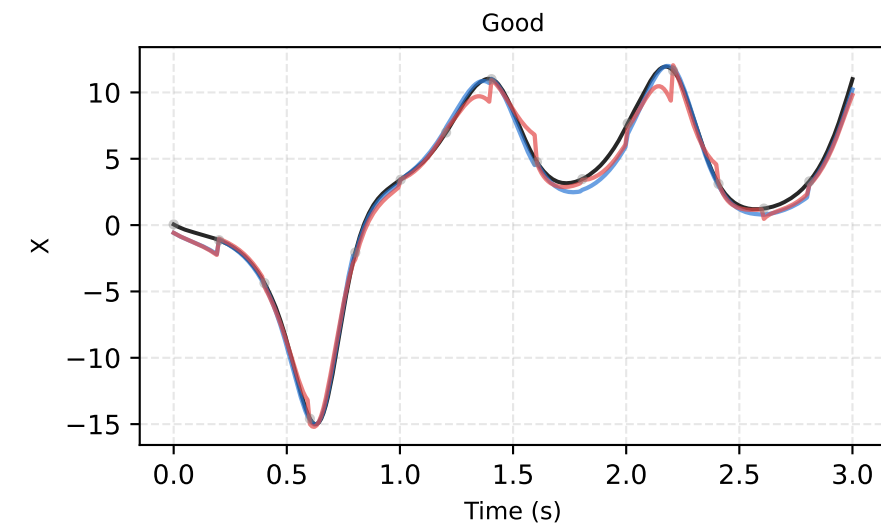
S0 — Weak-4DVar: Default vs. Joint (good / median / worst reconstructions)



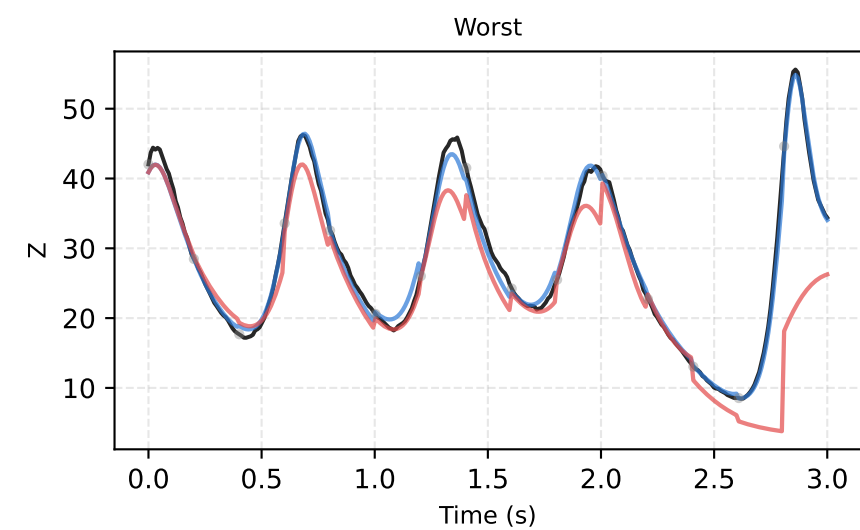
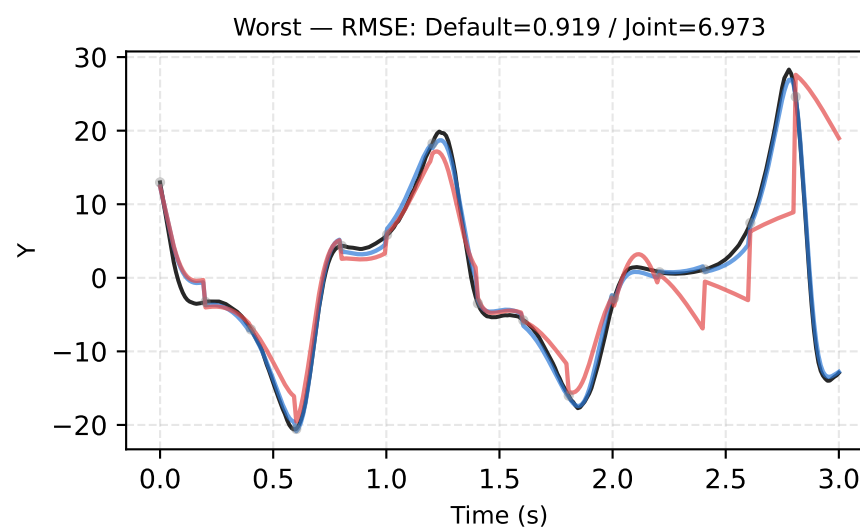
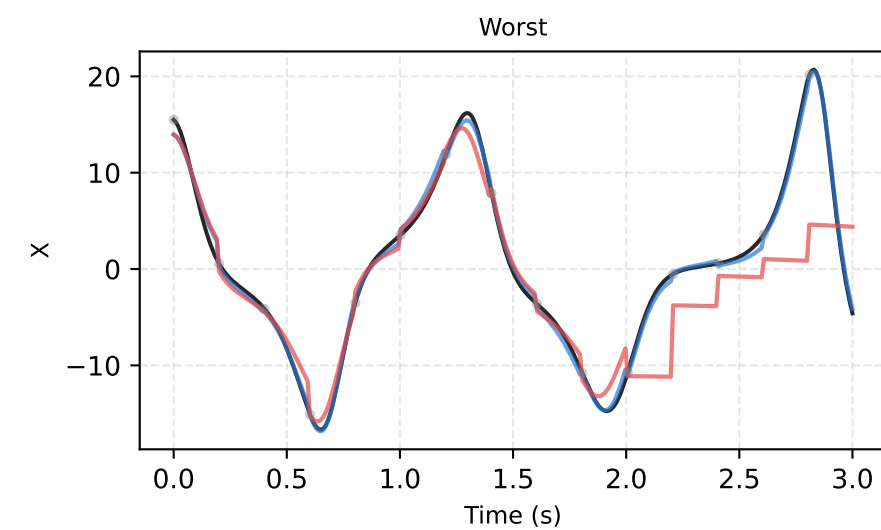
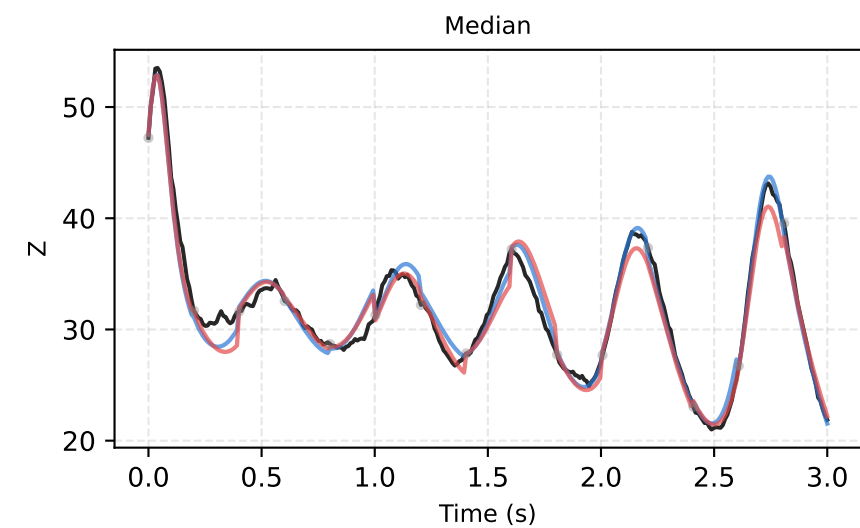
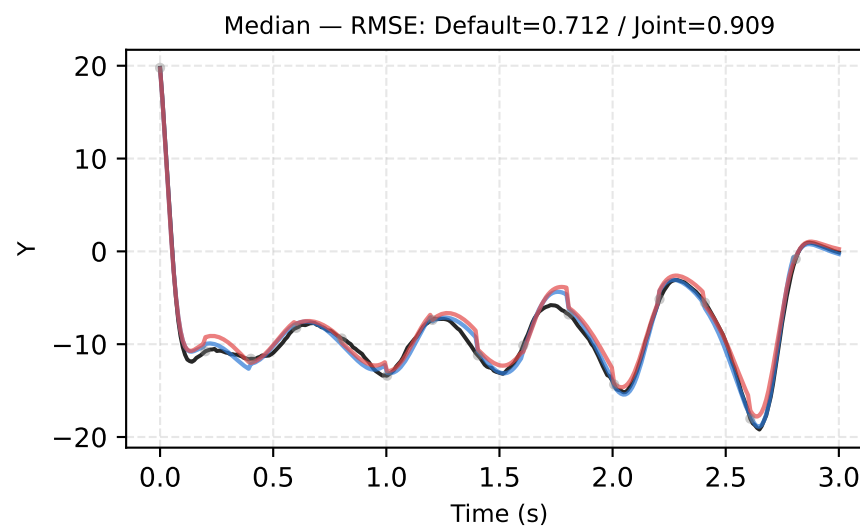
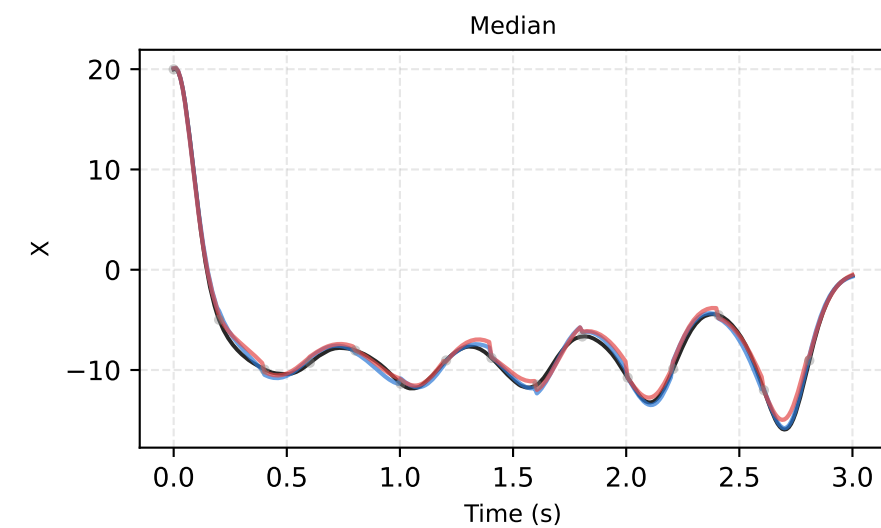
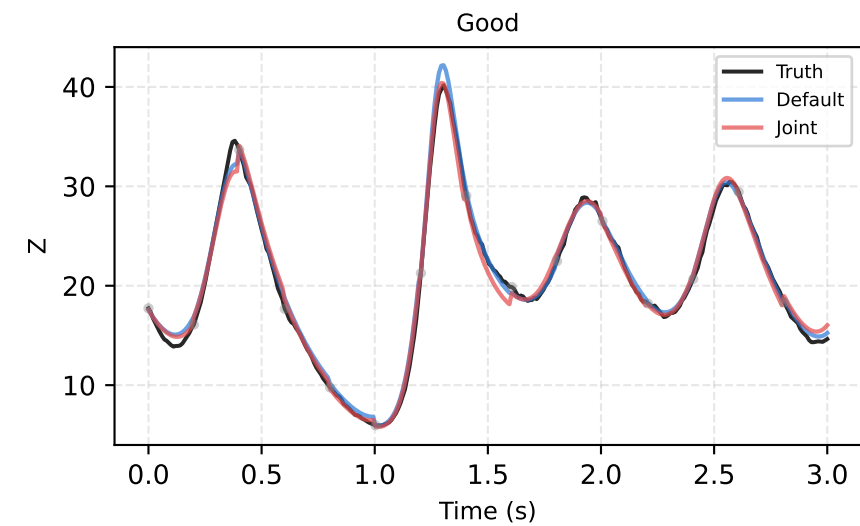
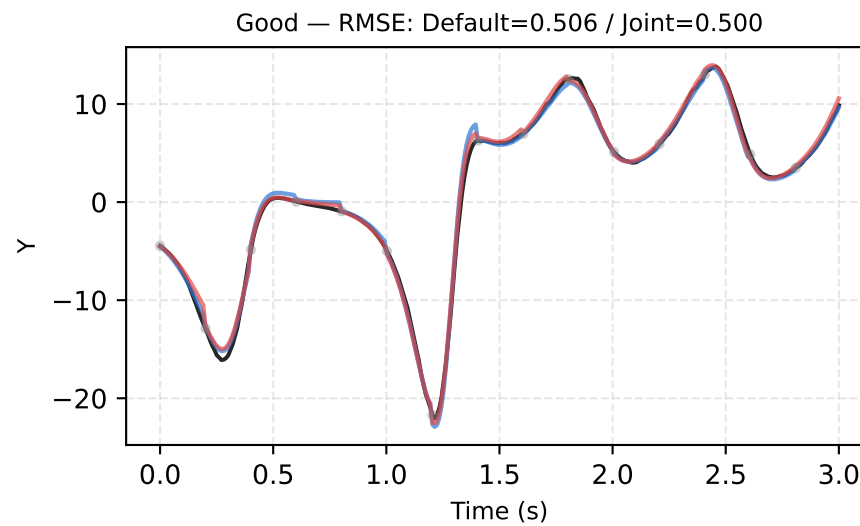
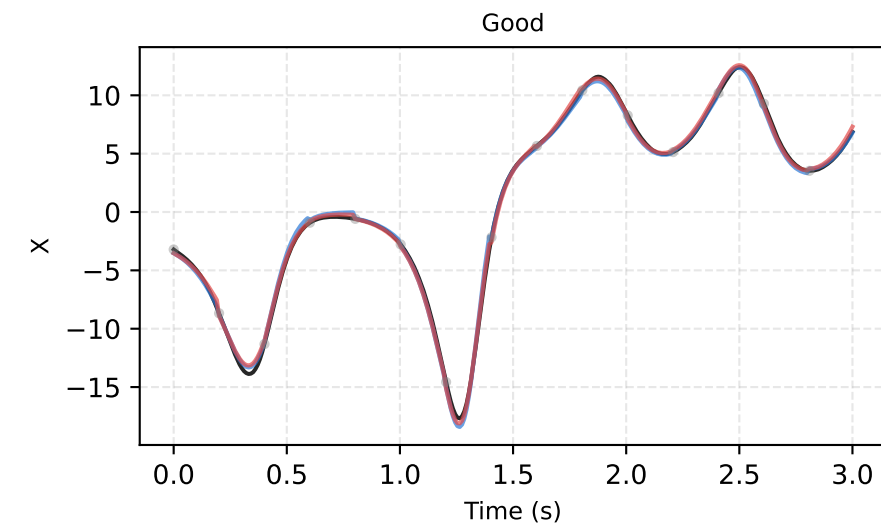
S0 — Strong-4DVar: Default vs. Joint (good / median / worst reconstructions)



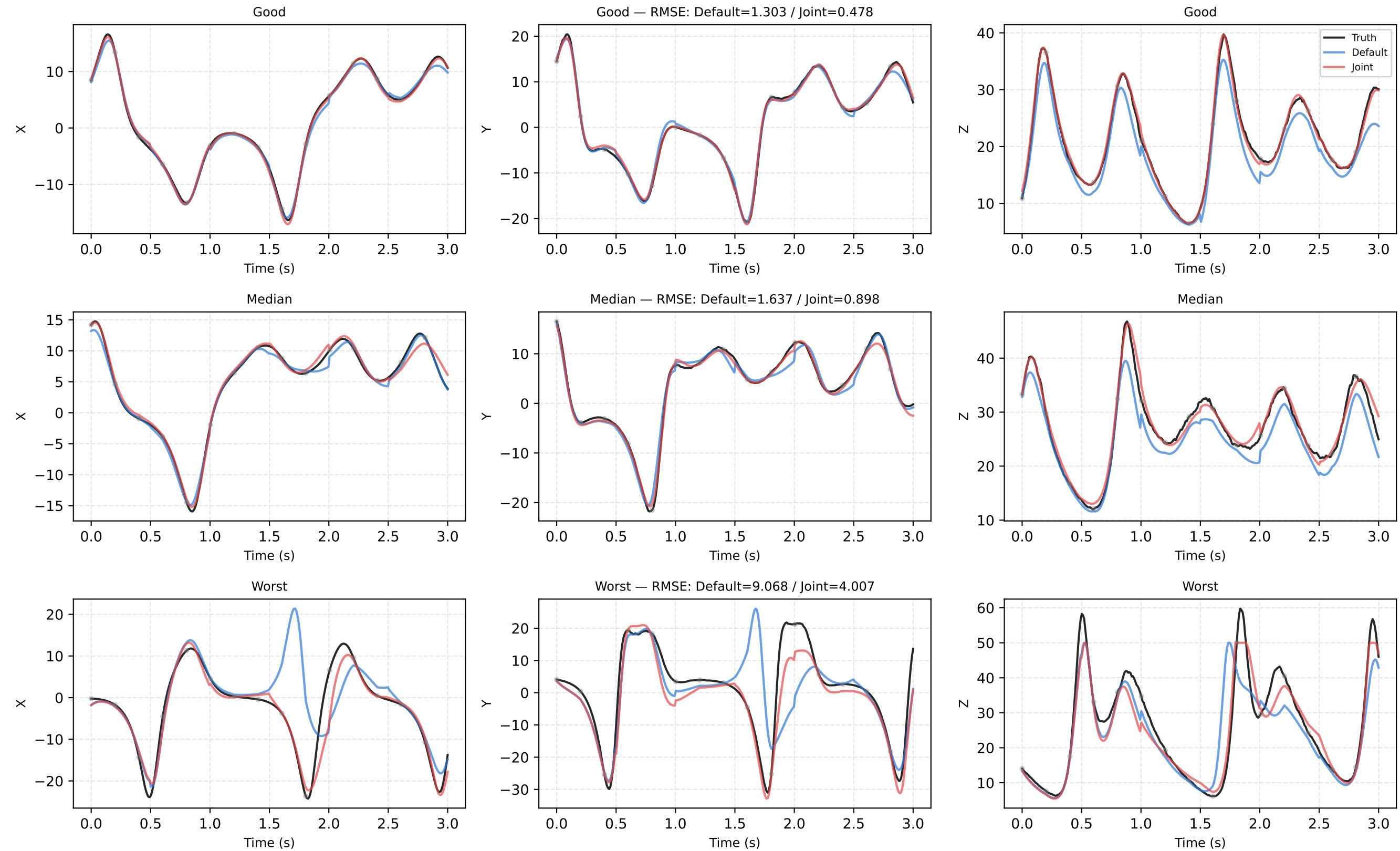
S0 — EnKF: Default vs. Joint (good / median / worst reconstructions)



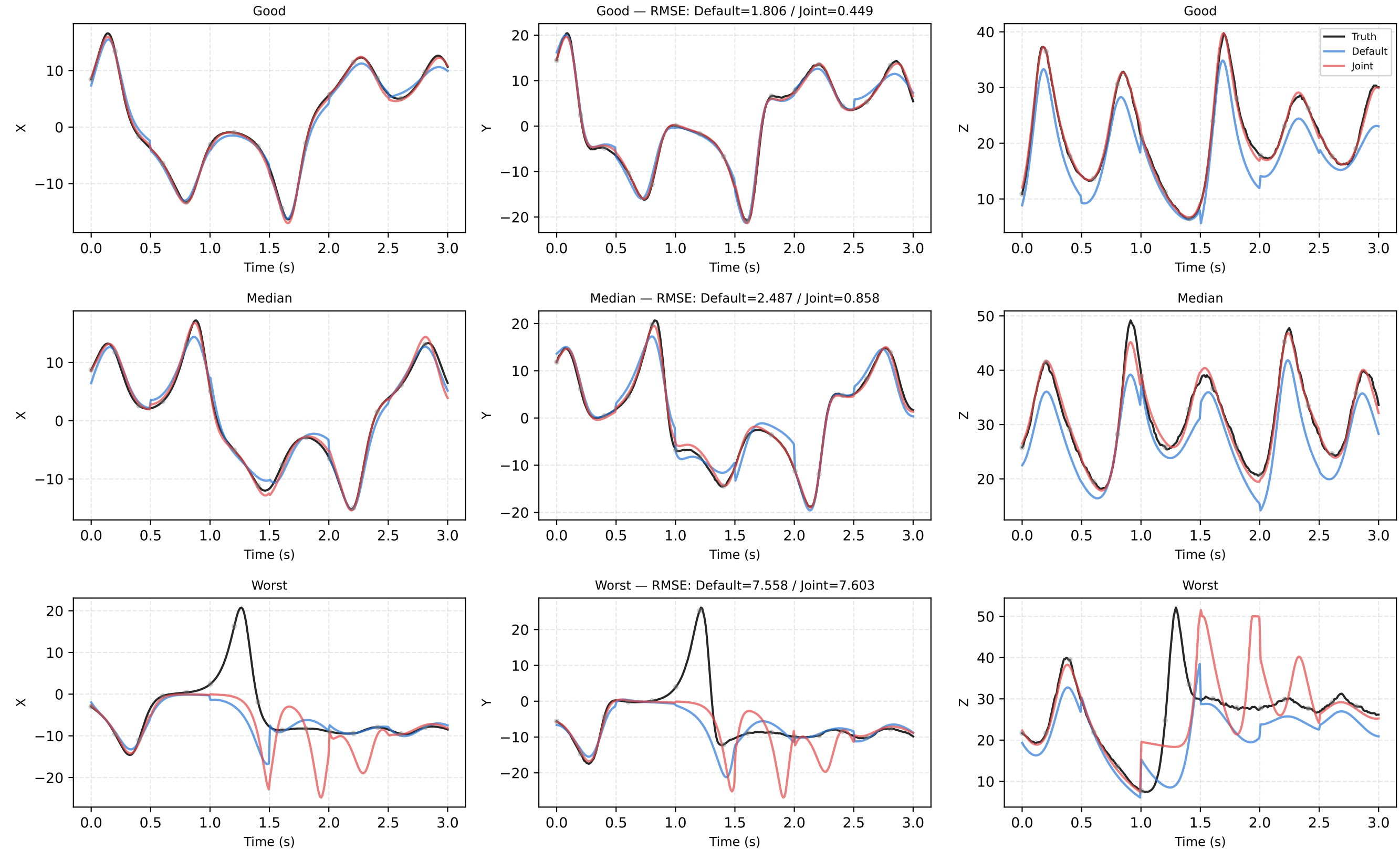
S0 — ETKF: Default vs. Joint (good / median / worst reconstructions)



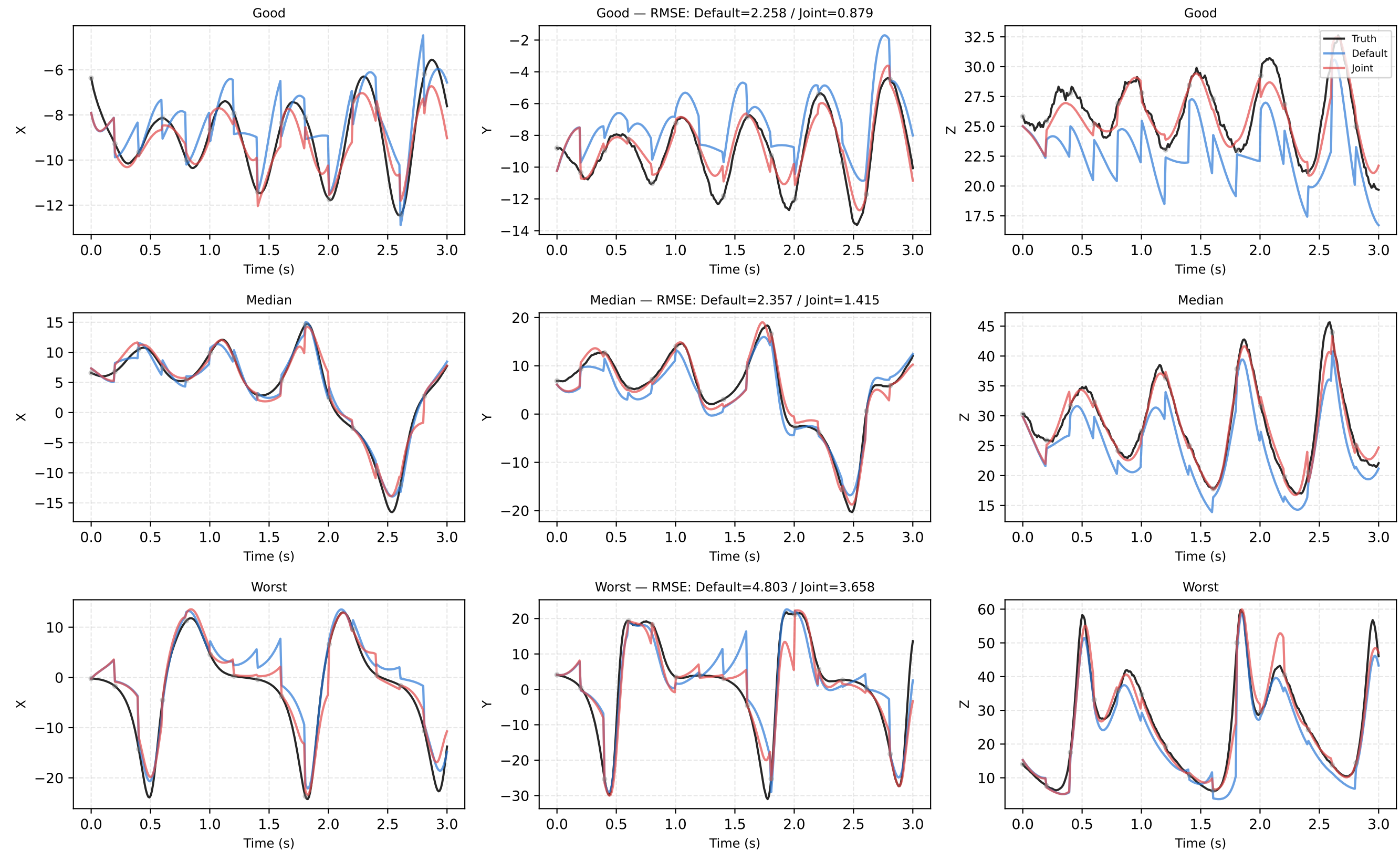
S1 — Weak-4DVar: Default vs. Joint (good / median / worst reconstructions)



S1 — Strong-4DVar: Default vs. Joint (good / median / worst reconstructions)

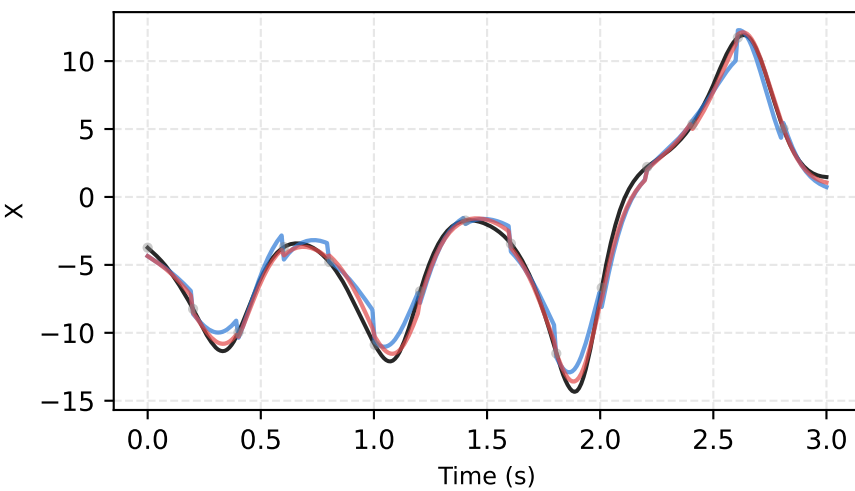


S1 — EnKF: Default vs. Joint (good / median / worst reconstructions)

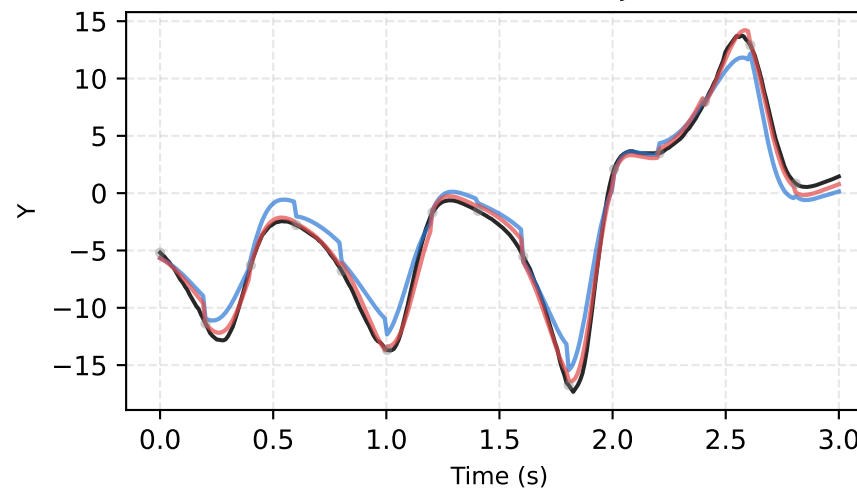


S1 — ETKF: Default vs. Joint (good / median / worst reconstructions)

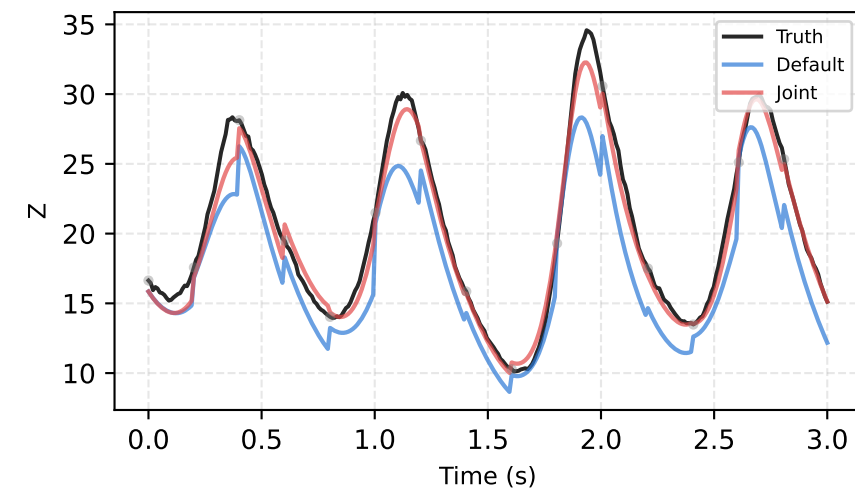
Good



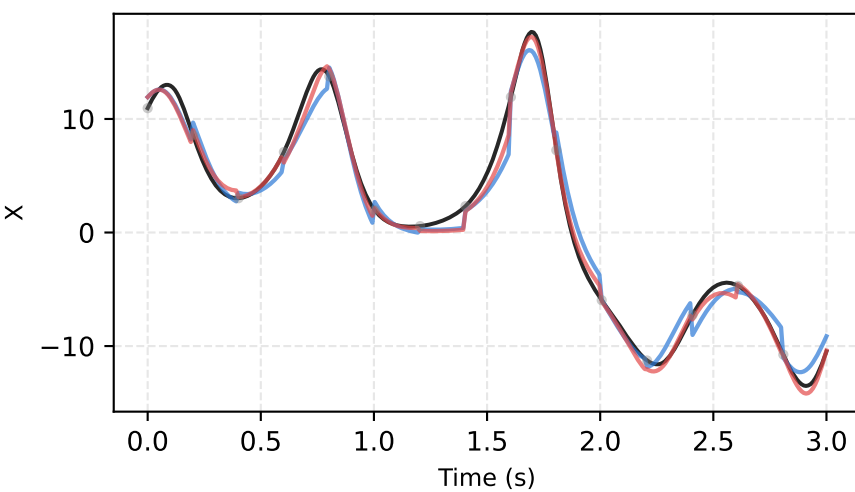
Good — RMSE: Default=1.814 / Joint=0.705



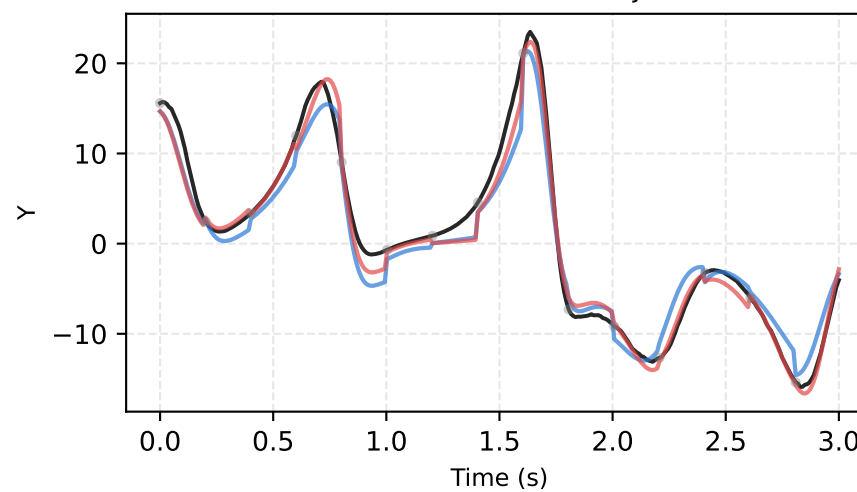
Good



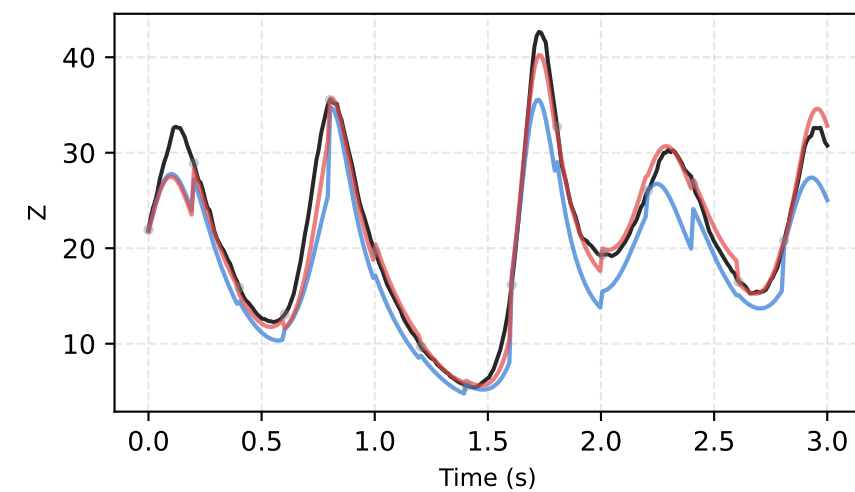
Median



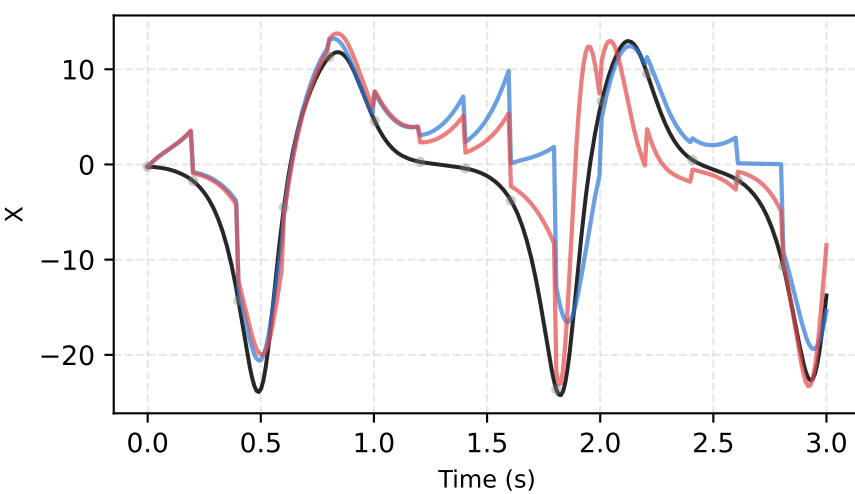
Median — RMSE: Default=2.319 / Joint=1.325



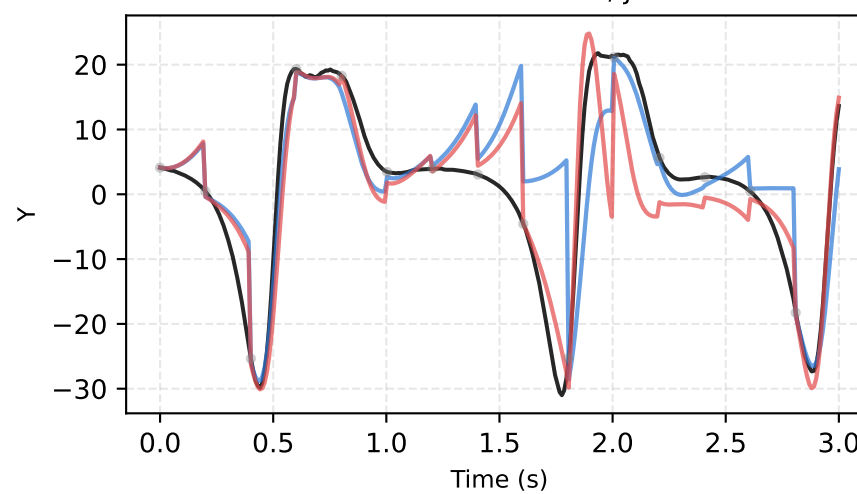
Median



Worst



Worst — RMSE: Default=6.773 / Joint=5.547



Worst

